

Machine Learning Project: To bee or not to bee

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Contributions

- Lianghong LI: project lead; implemented feature extraction, model training, validation, final prediction pipeline, and report integration.
- Zhekai YAN: support contribution for data checking, visualization review, and report proofreading.
- Muzi LI: support contribution for reproducibility checking, documentation review, and result verification.

Introduction

This project classifies insect images into the assignment target bug type. The main pipeline uses handcrafted features extracted from each image and its segmentation mask, not raw-image CNN classification. The mask lets the pipeline measure foreground color, shape, and texture while excluding background pixels.

Dataset and Preprocessing

The training labels contain ID, bug type, and species. The supervised target is bug type; species is used only for visualization and descriptive comments. The official test set contains 97 images with IDs 251-347 and corresponding masks.

ID 154 is excluded because train/masks/binary_154.tif is unavailable. No artificial, estimated, manually drawn, or imputed mask was created. The effective mask-based training set therefore contains 249 samples.

Effective training samples: 249.

bug type	count
Bee	115
Bumblebee	100
Butterfly	15
Hover fly	9
Wasp	9
Dragonfly	1

species	count
Bombus hortorum	71
Apis mellifera	58
Bombus pascuorum	25
Anthidium manicatum	19
Megachile centuncularis	17
Eristalis	9
Vespula germanica	8
Anthidium	6
Macroglossum stellatarum	5
Andrenidae	4

Feature Extraction

All mask-based statistics exclude pixels outside the segmentation mask. Masks are converted to grayscale and binarized with $\text{mask} > 0$, so every feature is derived from foreground insect pixels or the binary foreground shape.

- Color symmetry compares the insect bounding-box crop with its left-right flipped version. RGB differences are measured only where both original and flipped masks contain foreground pixels.
- Shape symmetry flips the binary mask crop left-right and measures normalized foreground overlap with the original crop.
- Bug pixel ratio is $\text{bug_pixels} / \text{image_pixels}$, capturing the relative size of the segmented insect in the photograph.
- RGB features are min, max, mean, median, and standard deviation for R, G, and B inside the mask.
- Morphology features include area, perimeter, circularity, aspect ratio, extent, solidity, orientation, eccentricity, major/minor axis lengths, and equivalent diameter.
- Hu moments provide compact global shape descriptors that summarize mask geometry beyond simple size and axis measurements.
- HSV and LAB summaries complement RGB by separating hue, saturation, value, lightness, and perceptual color axes.
- Grayscale contrast and edge density describe texture and local structure inside the mask, which can help when bug types share similar colors.

Data Visualization

The bug type distribution shows strong class imbalance, which motivates macro-F1. The species distribution and species-by-bug-type plot show that species labels are finer-grained and imbalanced; species is therefore kept descriptive and is not used as the final target.

PCA gives a linear view with partial separation and substantial overlap. t-SNE shows local grouping but should not be over-interpreted globally. The isomap projection checks nonlinear structure; it reveals useful geometry, but not perfectly separated classes.

Figure 1. Bug type distribution

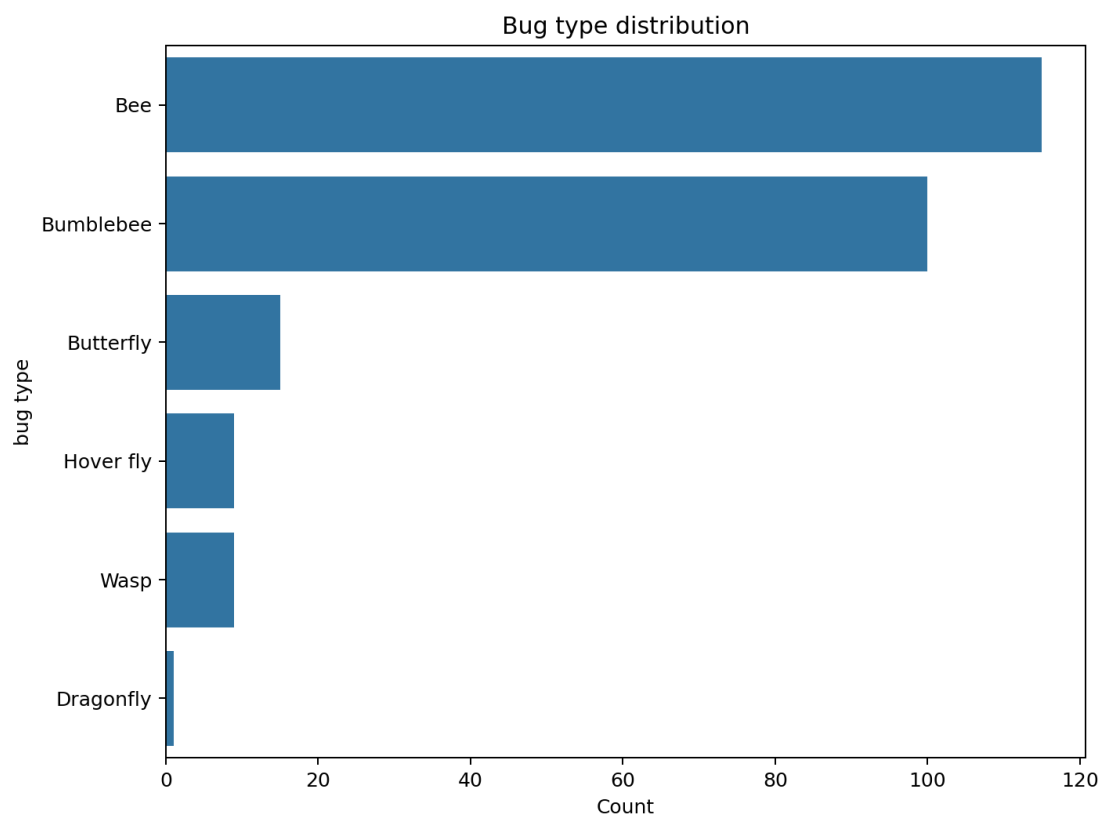


Figure 2. Species distribution

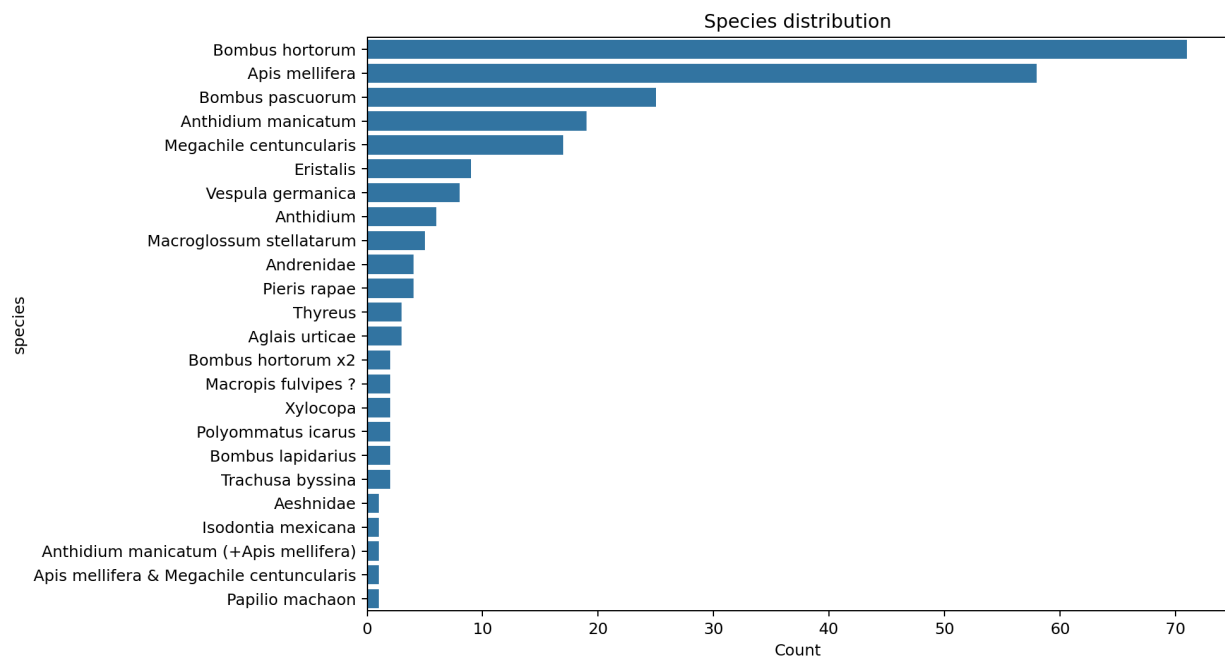


Figure 3. Species grouped by bug type

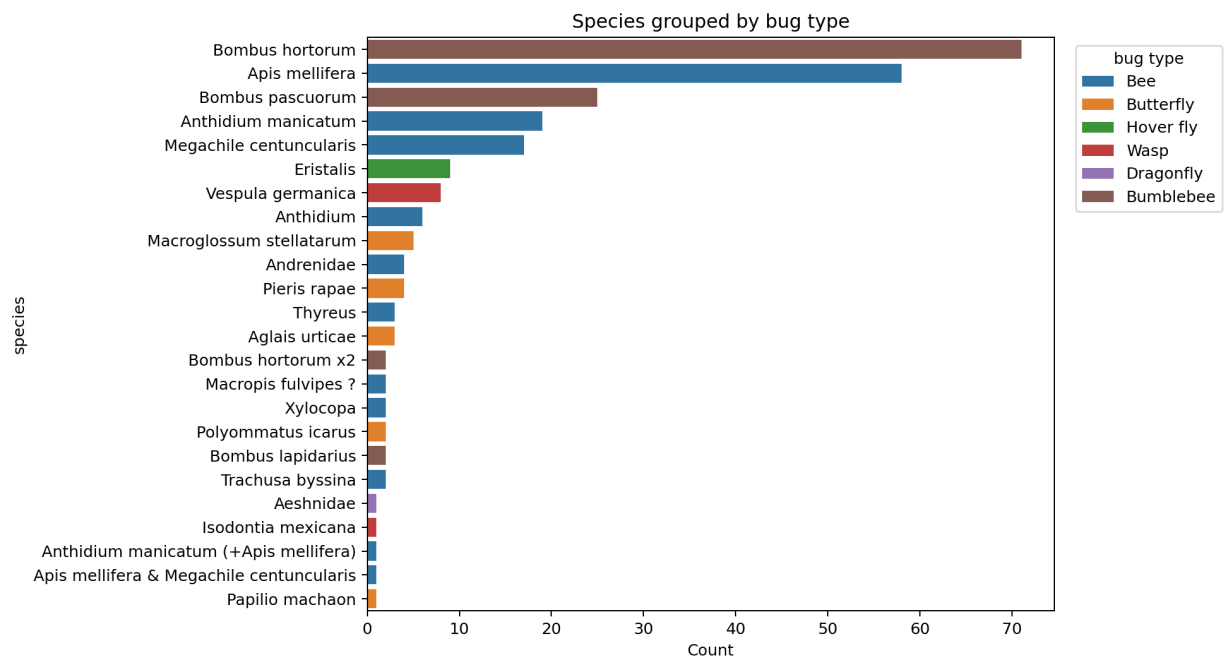


Figure 4. PCA 2D projection colored by bug type

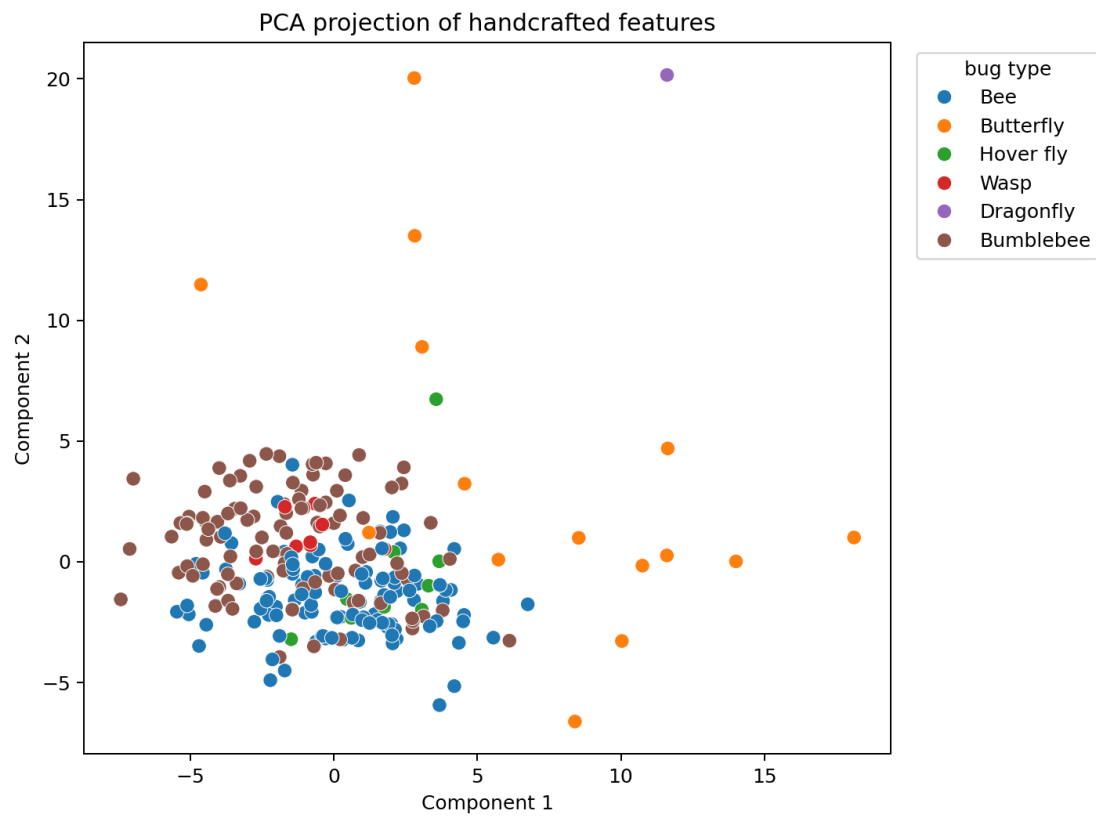


Figure 5. t-SNE 2D projection colored by bug type

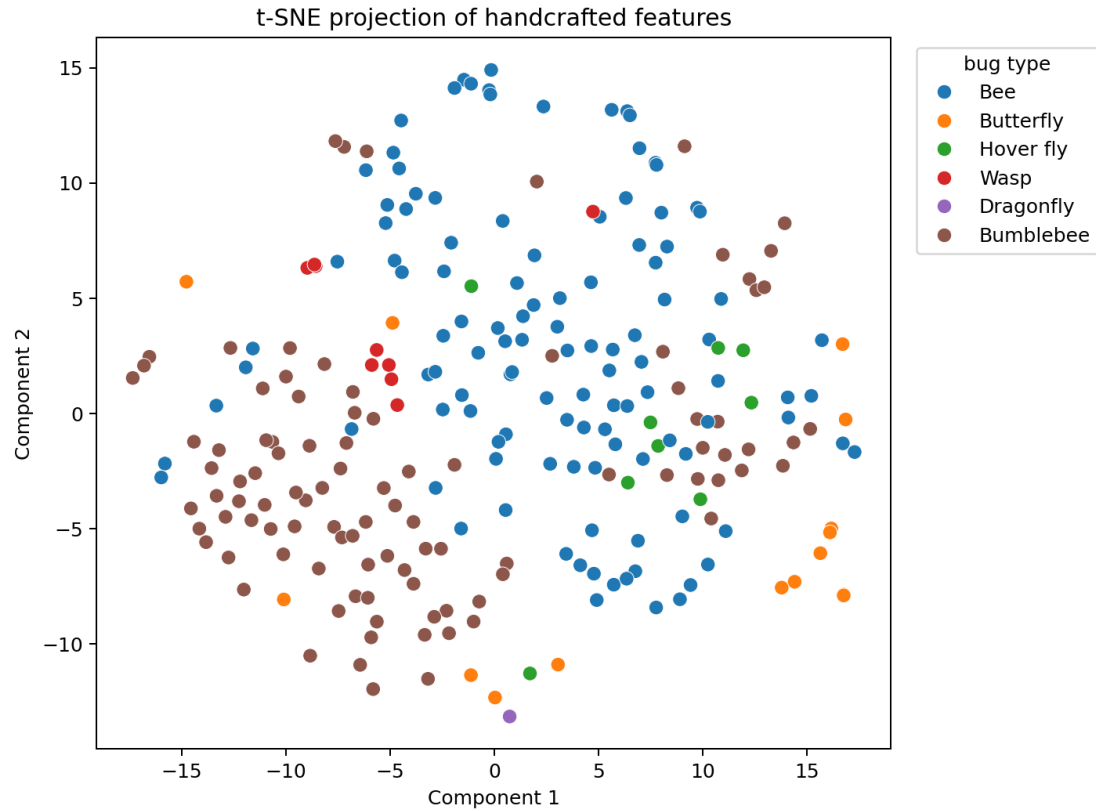
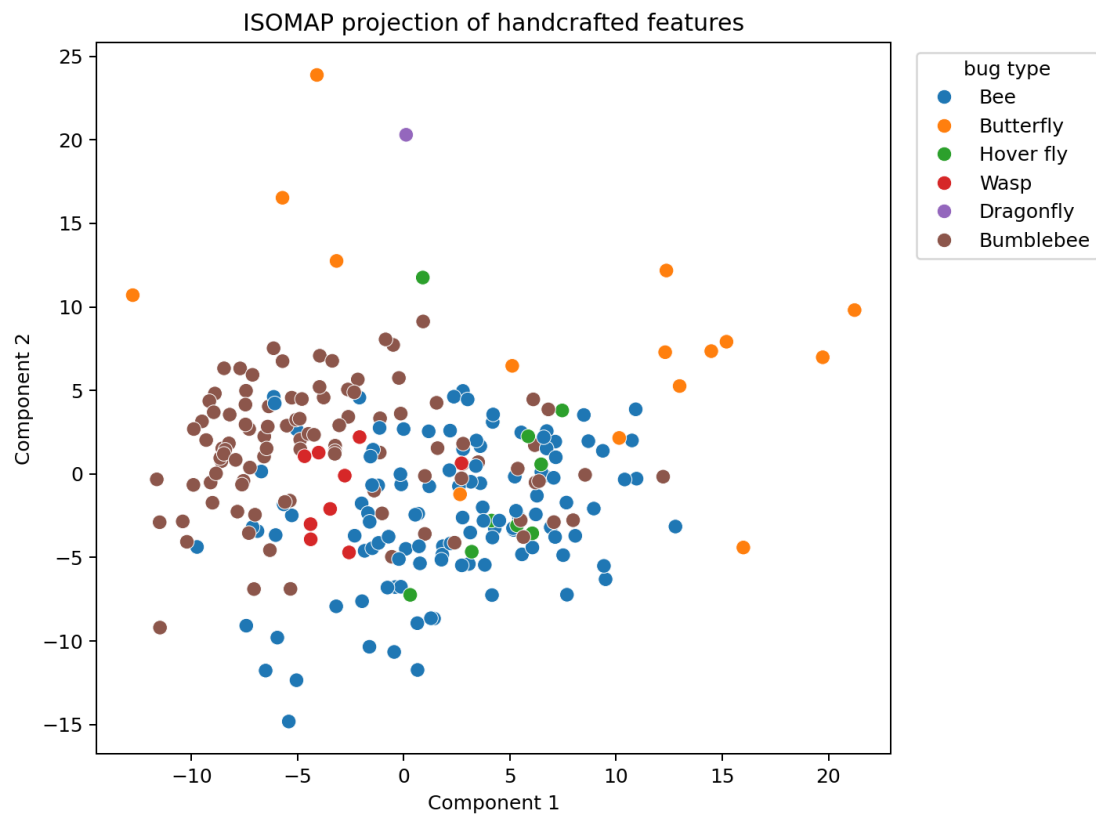


Figure 6. ISOMAP 2D projection colored by bug type



Supervised Learning Methods

The supervised target is bug type, never species. Handcrafted image/mask features are used for the main task because the assignment requires a feature-based pipeline. Macro-F1 is the main validation metric because the dataset is imbalanced. Frequent-class cross-validation is used because singleton classes make full stratified CV invalid. Full-data metrics are descriptive checks only and are not validation metrics.

Frequent-class CV used 5 folds and excluded only the rare classes that could not support stratification. The final selected model is then refit on all 249 usable training samples, including rare classes.

- DummyClassifier is the majority-class baseline.
- LogisticRegression is a linear non-ensemble baseline with class balancing.
- SVM RBF models smooth nonlinear boundaries in standardized tabular feature space.
- KNN tests whether local feature neighborhoods agree with labels.
- RandomForest, ExtraTrees, and GradientBoosting provide supervised ensemble comparisons.
- MLP is a shallow neural model trained on extracted features only, not raw images.

model	accuracy	macro_f1	weighted_f1	status
mlp	0.871	0.755	0.868	ok
svm_rbf	0.863	0.754	0.859	ok
logistic_regression	0.851	0.698	0.856	ok
extra_trees	0.847	0.618	0.827	ok
gradient_boosting	0.798	0.597	0.790	ok
knn	0.786	0.583	0.766	ok
random_forest	0.819	0.564	0.796	ok
dummy_most_frequent	0.464	0.127	0.294	ok

Clustering Methods

method	clusters	noise	silhouette	ARI	NMI
kmeans	6.000	0.000	0.113	0.149	0.235
agglomerative	6.000	0.000	0.077	0.076	0.181
dbscan	0.000	249.000		0.000	0.000

KMeans gives the strongest but still low/moderate clustering scores. AgglomerativeClustering finds imperfect hierarchical structure. DBSCAN marks all samples as noise under the selected settings. These results indicate that the handcrafted features contain useful structure but do not form perfectly separated natural bug type clusters.

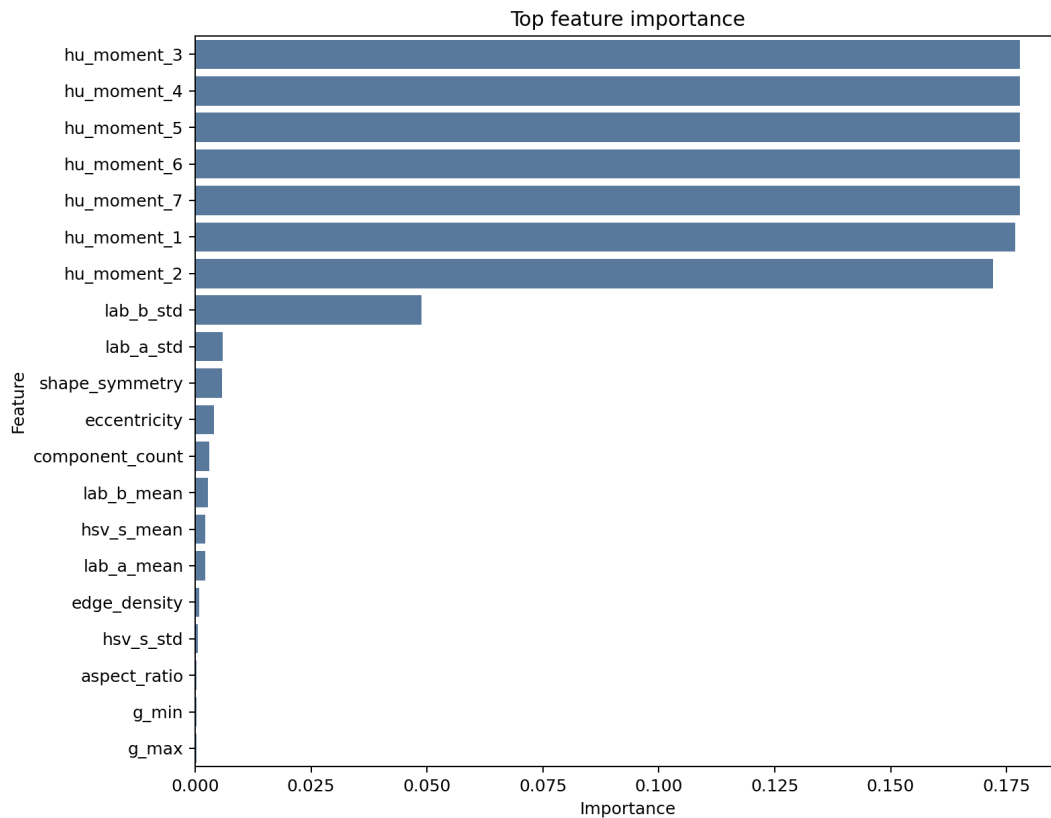
Model Selection

The metric table is kept unchanged and transparent: MLP has the numerically highest frequent-class CV macro-F1, while SVM RBF is selected for final prediction by a robustness tie-breaker.

- Numeric best model: mlp with CV macro-F1 0.7553521749357212.
- Final recommended model: svm_rbf with CV macro-F1 0.7535392750651922.
- The macro-F1 difference is below 0.005.
- Tie-breaker reason: SVM RBF selected by robustness tie-breaker because its macro-F1 is within 0.005 of MLP and it is simpler/stabler for small imbalanced data.

SVM RBF is preferred because it is simpler and more stable for a small imbalanced handcrafted-feature dataset while performing essentially the same as MLP in validation.

Figure 7. Descriptive permutation importance for the final fitted model



Final Prediction on Test Images 251-347

The official test images and masks for IDs 251-347 were processed with the same feature schema as training. The final SVM RBF model was applied to all 97 test samples.

results/submission.csv and deliverables/submission.csv were generated with exactly two columns: ID and bug type. Both files were validated for 97 rows, IDs 251-347 in order, no duplicates, no missing values, and valid training bug type labels.

predicted bug type	count
Bee	46
Bumblebee	38
Butterfly	6
Wasp	5
Hover fly	2

No test accuracy is reported because official test labels are unavailable. No predictions were manually edited or fabricated.

Limitations

The dataset is small and strongly imbalanced. The Dragonfly class has only one usable training sample. ID 154 is excluded because the official mask is missing. Bee and Bumblebee can visually overlap, making separation difficult for handcrafted features. Test labels are unavailable, so no test accuracy can be reported.

Conclusion

The project satisfies the required feature extraction, visualization, supervised learning, clustering, and final prediction components. Handcrafted mask-based features provide a compliant and interpretable representation, and SVM RBF is selected as the robust final model for the generated submission.

Reproducibility

- `python3 -m src.data_loader --validate --split train`
- `python3 -m src.build_features --split train`
- `python3 -m src.train_models`
- `python3 -m src.clustering`
- `python3 -m src.visualize`
- `python3 -m src.make_report`
- `python3 -m src.data_loader --validate --split test`
- `python3 -m src.build_features --split test`
- `python3 -m src.predict`
- `python3 -m src.validate_submission results/submission.csv`
- `python3 -m src.validate_submission deliverables/submission.csv`

Test data is not used for model selection or hyperparameter tuning.